

Toward Real-time Mobile Light Field Displays: An Efficient 3DGS Rendering Framework

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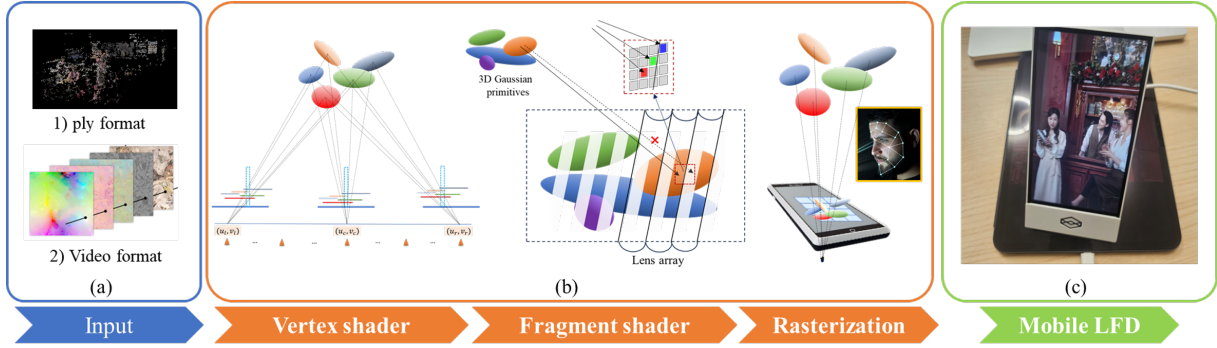


Figure 1: Overview of the system design for mobile light field experiences.

Abstract

This paper presents a real-time light field rendering framework for mobile devices based on 3D Gaussian Splatting (3DGS). The system is implemented using Apple’s *Metal* API, leveraging vertex amplification and subpixel mapping for efficient multi-view synthesis. Experiments on Apple M4 devices (MacBook Pro and iPad Pro) demonstrate real-time to interactive performance with eight viewpoints at a resolution of 1440×2560 . These results highlight the feasibility of 3DGS-based light field rendering on mobile platforms and the potential for practical handheld 3D experiences.

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1 Introduction

Light field displays can provide viewers with autostereoscopic vision with motion parallax, but real-time rendering remains challenging as dozens of views must be synthesized from limited references. 3D Gaussian Splatting (3DGS) [Kerbl et al. 2023] has emerged as an efficient alternative to conventional novel view synthesis methods, combining high quality with fast tile-based parallel rasterization of pre-trained Gaussian primitives. Nevertheless, when applied to light field rendering, the computation still scales linearly with the number of views, making further acceleration strategies necessary.

Prior studies have explored real-time view synthesis and multiplexing for large-scale light field displays, aiming to generate and combine input views at interactive rates. One line of work introduces tailored projective mappings [Wetzstein et al. 2011] that directly map display pixels to their corresponding rays, eliminating redundant multi-view rendering and significantly improving efficiency without compromising visual fidelity. Building upon this idea, DirectL [Peng et al. 2022] extends radiance field rendering to light field displays by designing ray-order pipelines for NeRFs and 3DGS with subpixel repurposing, achieving significant speedups while maintaining high-quality results.

On the implementation side, MetalSplat [Cier 2024] provides an open-source Metal implementation of 3D Gaussian Splatting, demonstrating the feasibility of high-performance rendering on

Apple GPUs. However, scaling to the higher view counts required for autostereoscopic light field displays remains an open challenge.

Therefore, this paper presents a 3DGS-based multi-view rendering framework optimized toward mobile light field displays, synthesizing up to eight simultaneous viewpoints through vertex amplification and subpixel mapping on Metal-based mobile devices.

2 System Design

To realize 3D light field reproduction on mobile devices, it is essential to design an efficient rendering pipeline that maps 3D Gaussian primitives into multiple view-dependent images. Figure 1 provides an overview of the entire process. Figure 1(a) illustrates the input stage, where pre-trained 3D Gaussian primitives obtained from a 3DGS reconstruction are provided in either PLY format or as video-based sequences. Figure 1(b) depicts the rendering pipeline that performs real-time multi-view synthesis using Metal on Apple M4 devices. The vertex shader amplifies each Gaussian primitive into multiple view instances, while the fragment shader rasterizes them at subpixel resolution in accordance with the lenticular-lens specifications, aligning each fragment with its corresponding directional ray.

To implement the proposed system, we employed two types of devices. First, the Looking Glass Go [Looking Glass Factory 2024] was used as a mobile light field display, as shown in Figure 1(c). Second, Apple devices equipped with the M4 chip—namely, a MacBook Pro and an iPad Pro—were utilized as rendering platforms to perform real-time synthesis from the given 3D Gaussian primitives. The details of our real-time light field rendering method based on 3D Gaussian Splatting are described in the following section.

3 Method

To accelerate real-time light field rendering based on 3D Gaussian primitives, we implemented the system in Metal to fully utilize the hardware acceleration of mobile GPUs. In the vertex shader, the *vertex amplification* feature extends a single draw call into multiple viewport instances. Each instance retrieves its viewport index from `amplification_id` and references the corresponding view–projection matrix. Each Gaussian’s 3D covariance is projected into screen space using the Jacobian of the view–projection transform, forming an elliptical footprint following the Elliptical Weighted Average (EWA) formulation.

In the fragment shader, a subpixel-mapping LUT, defined by the light field display’s resolution, assigns each fragment to its target viewport. Fragments outside the assigned view are discarded, while valid fragments proceed to splat accumulation. Framebuffer fetch enables the shader to access existing color and depth values for blending within the same pixel tile, supporting efficient subpixel-level updates under the tile-based deferred rendering (TBDR) architecture.

Each fragment is accumulated by standard α -blending for both color and depth:

$$X_{\text{out}} = X_{\text{prev}} \cdot (1 - \alpha) + X_{\text{new}} \cdot \alpha, \quad X \in \{C, Z\}. \quad (1)$$

Here, C denotes color and Z depth. Applying the same α -blending yields a continuous, alpha-weighted depth buffer that integrates

Table 1: Rendering performance (ms) at different view counts, measured at 1440×2560 resolution with 57 K Gaussian primitives on Apple M4 devices. Baseline: single-view MetalSplat-ter without proposed optimizations.

Device	Baseline	4 Views	8 Views
MacBook Pro (M4)	7.63	8.63	29.31
iPad Pro (M4)	14.15	45.61	78.45

semi-transparent Gaussians for smoother reprojection. Finally, using Metal’s ImageBlock tile memory ensures efficient per-pixel accumulation while preserving the blending semantics of Equation (1).

4 Results

Rendering performance was evaluated on a MacBook Pro (M4) and an iPad Pro (M4) at a resolution of 1440×2560 with 57 K Gaussian primitives. Frame times were measured using Metal GPU timestamps, representing pure GPU rendering duration independent of display vsync. For comparison, we used the original MetalSplat-ter [Cier 2024] as a single-view baseline, which renders without vertex amplification or subpixel mapping and uses only zeroth-order spherical harmonics (SH_0).

Table 1 summarizes the results. On the MacBook Pro (M4), the proposed system achieved average rendering times of 8.63 ms for 4 views and 29.31 ms (≈ 34 fps) for 8 views. On the iPad Pro (M4), rendering times were 45.61 ms for 4 views and 78.45 ms (≈ 13 fps) for 8 views, compared to the single-view baseline of 14.15 ms (≈ 71 fps). Experiments were limited to eight views due to Metal’s vertex amplification constraint.¹

These results demonstrate that GPU-level parallelization combined with Metal-specific optimizations—vertex amplification, ImageBlock tile memory, and the unified memory architecture—enables real-time multi-view rendering on laptop-class hardware and interactive rates on mobile devices. This establishes the feasibility of 3DGS-based light field rendering on mobile platforms, paving the way for practical portable light field displays.

5 Conclusion and Future Work

In this paper, we presented a real-time light field rendering framework based on 3D Gaussian Splatting (3DGS) for mobile environments and validated its performance through GPU optimization using the Metal API. Experiments on the MacBook Pro (M4) and iPad Pro (M4) demonstrated that the proposed framework scales efficiently to handle multiple views simultaneously and achieves real-time rendering on mobile hardware for a limited number of views. These results confirm the feasibility of implementing light field displays on mobile platforms. Future work includes extending beyond the current eight-view constraint, further performance optimization through adaptive-resolution rendering and video-decoding integration, and exploration of alternative Metal-based rendering strategies toward a complete and practical mobile light field pipeline.

¹On the M4 GPU (Apple9 family), the Metal API supports up to eight vertex amplification instances per draw call.

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